20. Robbins-Monroe

March 2, 2026 8:52 AM

Stochastic Approximation 1950's

Robbins - Monroe

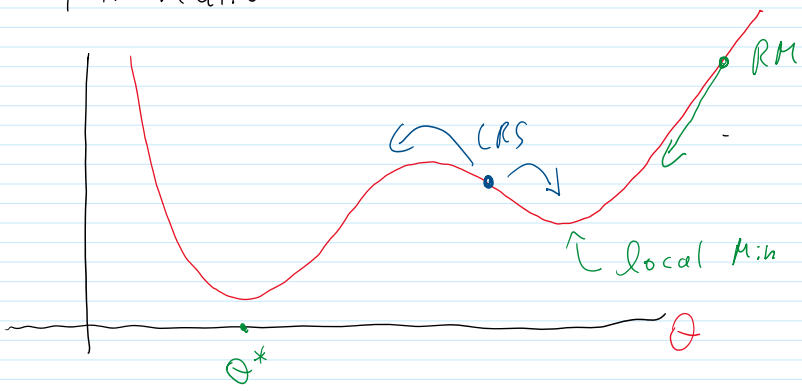
Like last week, our aim is to minimize some loss function $L(\theta)$, $\theta \in \mathbb{R}^d$ our model parameters.

Methods SRS, LRS, ELRS from last week are simple to run, not overly efficient, but can apply to non-convex problems.

concave

This week, Robbins Monroe (RM) and Kiefer-Wolfowitz (KW)

- These are Gradient based (related to Newton-Raphson)
- They are often faster for (noisy) convex optimization but get stuck in local minima for non-convex optimization.



- LRS may choose any direction
- RM only follows the gradient

Goal: Minimize $L(\theta)$

- This occurs when $\underbrace{\frac{d}{d\theta} L(\theta) = 0}_{\text{root Finding!}}$ (assuming derivative exists)
- If $L(\theta)$ were deterministic, then we could try NR or other related Methods

But NR will fail when the loss is noisy.

Setup: $L(\theta) = \mathbb{E}[Q(\theta, v)]$

Loss

Noisy loss

parameters

Noise

Loss

Noisy Loss parameters

We don't get to see $\frac{d}{d\theta} L(\theta)$ ← true gradient

Instead we get $y(\theta) = \frac{\partial}{\partial \theta} Q(\theta, v)$

Noisy Gradient

$\boxed{\text{RM:}}$ Given some θ_k, kth step of the Algorithm

Then $\theta_{k+1} = \theta_k - a_k \, y(\theta_k)$

new = old − (#) · (Noisy Gradient)

NR $x_{k+1} = x_k - \frac{L(x_k)}{L'(x_k)}$

→ what is a_k? "Gain Sequence", $\boxed{a_k > 0}$

a_k are step-sizes, How far to move in the direction of the gradient,

- we want $a_k \to 0$ so RM stops eventually
- we also want it to shrink slowly enough so that we explore the θ's well enough

$\Rightarrow \sum_{k=1}^{\infty} a_k = \infty$, $\sum_{k=1}^{\infty} a_k^2 < \infty$

example: $a_k = \frac{1}{k}$ ✓

Sample Mean, $\bar{X}$, is a Stochastic Approximation!

- if we have data coming in $X_1, X_2, X_3, \ldots, X_n, \ldots$ s.t. X_n are iid with Mean $\mu = \mathbb{E}X_1$
- Problem: estimate the mean
- Solution: $\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$

Why $\bar{X}_n$? Because it Minimizes the Squared error!

Revisit $\bar{X}$

new observation ↓ old observation ↓

Revisit $\bar{X}$

new observation

old observations

$$\bar{X}_{n+1} = \frac{1}{n+1} X_{n+1} + \frac{1}{n+1} \frac{\sum_{i=1}^{n} X_i}{n} \, n$$

$$= \frac{1}{n+1} X_{n+1} + \frac{n}{n+1} \bar{X}_n$$

$$= \frac{1}{n+1} X_{n+1} + \left(1 - \frac{1}{n+1}\right) \bar{X}_n$$

$$= \bar{X}_n - \frac{1}{n+1} \left(\bar{X}_n - X_{n+1}\right)$$

$$\theta_{n+1} = \theta_n - \frac{1}{n+1} y(\theta_n)$$

Here: $L(\theta) = \frac{1}{2}(\theta - \mu)^2$ ← No Noise, so $\theta^* = \mu$

$$L'(\theta) = \theta - \mu$$

Mean Zero

$$\underbrace{y(\theta_n)}_{\text{Noisy Loss Gradient}} = \theta_n - X_{n+1} = \underbrace{(\theta_n - \mu)}_{\text{Noise-Free Gradient}} + \underbrace{(\mu - X_{n+1})}_{\text{additive Noise}}$$